Project Design Phase
Proposed Solution Template

Date	14 February 2026
Team ID	LTVIP2026TMIDS81120
Project Name	Smart Sorting: Transfer Learning for Identifying Rotten Fruits and Vegetables
Maximum Marks	2 Marks

Proposed Solution Template — Fruit Freshness Classification System

To address the fruit freshness classification problem, a deep learning–based image classification approach is proposed using a convolutional neural network (CNN). The system analyzes uploaded fruit images by extracting visual features such as color variations, texture patterns, and surface characteristics to predict whether a fruit is Fresh or Spoiled.

A CNN is selected as the final model because of its high accuracy, efficiency, and strong capability in handling complex visual patterns. It automatically learns hierarchical image features without manual feature engineering. By applying convolution and pooling operations, the model identifies important visual cues that distinguish fresh fruits from spoiled ones, improving prediction reliability and reducing classification errors.

The trained CNN model is saved and integrated into a Flask-based web application. When users upload a fruit image through the interface, the system preprocesses the image and passes it to the trained model, which instantly generates a freshness prediction. This approach ensures fast, accurate, and reliable results, making the system suitable for real-world quality inspection and food safety applications.

How the CNN Works in the Fruit Freshness Prediction Model

The convolutional neural network works by progressively learning visual patterns from fruit images.

First, the model receives an input image and applies convolution layers to detect low-level features such as edges, shapes, and textures. These features are essential in identifying subtle differences between fresh and spoiled fruits.

Next, pooling layers reduce image dimensions while preserving important information, helping the model focus on significant patterns. As the network deepens, it learns more complex representations such as color distribution and surface irregularities.

This learning process continues through multiple layers, gradually improving classification accuracy. During training, the model compares predictions with actual labels and adjusts its internal weights to minimize error.

Finally, when a new image is uploaded through the web application, the trained CNN processes the image and outputs whether the fruit is Fresh or Spoiled, delivering a quick and dependable prediction.